

QUESTIONS 8 / 8 completed

Logger ▾

LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the interface in a class called "Log4jLogger" that uses the Log4j logging framework

PROGRAM EDITOR

Java ▾

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Logger logger = new Log4jLogger();
4         logger.debug("This is a debug message.");
5         logger.info("This is an info message.");
6         logger.warning("This is a warning message.");
7         logger.error("This is an error message.");
8     }
9 }
10 interface Logger {
11     void debug(String message);
12     void info(String message);
13     void warning(String message);
14     void error(String message);
15 }
16 class Log4jLogger implements Logger {
17     @Override
18     public void debug(String message) {
19         System.out.println("[DEBUG] " + message);
20     }
21     @Override
```

OUTPUT

```
[DEBUG] This is a debug message.
[INFO] This is an info message.
[WARNING] This is a warning message.
[ERROR] This is an error message.
```

INPUT

Your INPUT goes here!

QUESTIONS 8 / 8 completed

Character ▾

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of character, such as "Hero", "Villain", and "Monster".

PROGRAM EDITOR

Java ▾

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         Character hero = new Hero("Arthur");
4         Character villain = new Villain("Malakor");
5         Character monster = new Monster("Gorgon");
6         System.out.println("--- Game Start ---");
7         hero.move();
8         hero.attack();
9         hero.takeDamage(10);
10        System.out.println();
11        villain.move();
12        villain.attack();
13        villain.takeDamage(20);
14        System.out.println();
15        monster.move();
16        monster.attack();
17        monster.takeDamage(15);
18    }
19 }
20 interface Character {
21     void move();
```

OUTPUT

```
--- Game Start ---
Arthur the Hero moves forward bravely!
Arthur attacks with a shining sword!
Arthur took 10 damage but stays strong!

Malakor the Villain creeps through the shadows!
Malakor launches a dark magic spell!
Malakor took 20 damage and growls in anger!

Gorgon the Monster stomps loudly!
```

INPUT

Your INPUT goes
here!

QUESTIONS 8 / 8 completed

Sort ▾

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         List<Integer> numbers = new ArrayList<>(Arrays.asList(5, 2, 9, 1, 7));
5         Collections.sort(numbers);
6         System.out.println("Sorted in ascending order: " + numbers);
7     }
8 }
```

OUTPUT

```
Sorted in ascending order: [1, 2, 5, 7, 9]
```

INPUT

Your INPUT goes here!

QUESTIONS 8 / 8 completed

HashSet ▾

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.HashSet;
2 public class HashSetRemoveElement {
3     public static void main(String[] args) {
4         HashSet<String> set = new HashSet<>();
5         set.add("Apple");
6         set.add("Banana");
7         set.add("Cherry");
8         set.add("Mango");
9         System.out.println("Original HashSet: " + set);
10        String toRemove = "Banana";
11        boolean removed = set.remove(toRemove);
12        if (removed) {
13            System.out.println(toRemove + " removed successfully.");
14        } else {
15            System.out.println(toRemove + " not found in the HashSet.");
16        }
17        System.out.println("HashSet after removal: " + set);
18    }
19 }
```

OUTPUT

```
Original HashSet: [Apple, Cherry, Mango, Banana]
Banana removed successfully.
HashSet after removal: [Apple, Cherry, Mango]
```

INPUT

Your INPUT goes here!

QUESTIONS 8 / 8 completed

TreeMap ▾

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.TreeMap;
2 import java.util.Map;
3 public class TreeMapExample {
4     public static void main(String[] args) {
5         TreeMap<String, Integer> map = new TreeMap<>();
6         map.put("Apple", 50);
7         map.put("Orange", 30);
8         map.put("Banana", 20);
9         map.put("Grape", 40);
10        System.out.println("TreeMap Entries (Sorted by Key):");
11        for (Map.Entry<String, Integer> entry : map.entrySet()) {
12            System.out.println("Key: " + entry.getKey() + ", Value: " + entry.getValue());
13        }
14    }
15 }
```

OUTPUT

```
TreeMap Entries (Sorted by Key):
Key: Apple, Value: 50
Key: Banana, Value: 20
Key: Grape, Value: 40
Key: Orange, Value: 30
```

INPUT

Your INPUT goes here!

QUESTIONS 8 / 8 completed

PriorityQueue ▾

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.PriorityQueue;
2 import java.util.Scanner;
3 public class Main {
4     public static void main(String[] args) {
5         PriorityQueue<Integer> pq = new PriorityQueue<>();
6         Scanner sc = new Scanner(System.in);
7         if (sc.hasNextInt()) {
8             while (sc.hasNextInt()) {
9                 int num = sc.nextInt();
10                pq.add(num);
11            }
12        }
13        System.out.println("PriorityQueue elements: " + pq);
14        sc.close();
15    }
16 }
```

OUTPUT

```
PriorityQueue elements: [1, 2, 3, 4, 5]
```

INPUT

```
1
2
3
4
5
```

QUESTIONS 8 / 8 completed

LinkedListHashMap ▾

LINKEDHASHMAP

Write a Java program to create a LinkedListHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.LinkedListHashMap;
2 import java.util.Map;
3 public class Main {
4     public static void main(String[] args) {
5         LinkedListHashMap<String, Integer> map = new LinkedListHashMap<>();
6         map.put("Apple", 10);
7         map.put("Banana", 20);
8         map.put("Orange", 30);
9         map.put("Mango", 40);
10        System.out.println("Keys in insertion order:");
11        for (String key : map.keySet()) {
12            System.out.println(key);
13        }
14    }
15 }
```

OUTPUT

```
Keys in insertion order:
Apple
Banana
Orange
Mango
```

INPUT

Your INPUT goes here!

QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class MaxMinSet {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         Set<Integer> numbers = new HashSet<>();
6         int n = sc.nextInt();
7         for (int i = 0; i < n; i++) {
8             numbers.add(sc.nextInt());
9         }
10        if (numbers.isEmpty()) {
11            System.out.println("The set is empty.");
12        } else {
13            int max = Collections.max(numbers);
14            int min = Collections.min(numbers);
15            System.out.println("Set elements: " + numbers);
16            System.out.println("Maximum value: " + max);
17            System.out.println("Minimum value: " + min);
18        }
19        sc.close();
20    }
21 }
```

OUTPUT

```
Set elements: [1, 2, 3, 4, 5]
Maximum value: 5
Minimum value: 1
```

INPUT

```
5
1
2
3
4
5
```